e-ISSN: 2582-5208
International Research Journal of Modernization in Engineering Technology and Science
(Peer-Reviewed, Open Access, Fully Refereed International Journal)
Volume:06/Issue:03/March-2024 Impact Factor- 7.868 www.irjmets.com

ADVANCING YOGA PRACTICE: LEVERAGING OPENCV AND MEDIAPIPE FOR POSE DETECTION

Maddi Bhargavi[*1], Bade Vineela[*2], Majji Hemanth[*3], Shahukara Khushi Kumar[*4], Nadupuru Sai Sumanth[*5], Tammineni Ravi Kumar[*6]

[*1,2,3,4,5]B.Tech UG Students, Dept. Of CSE, Aditya Institute Of Technology And Management, Tekkali, AP, India.

[*6]Guide, Professor & Asst. Head Of Department, Dept. Of CSE, Aditya Institute Of Technology And Management, Tekkali, AP, India.

DOI : https://www.doi.org/10.56726/IRJMETS51313

ABSTRACT

Yoga is a centuries-old practice that has become more popular all over the world due to its benefits for the body, mind, and spirit[1]. Nowadays, yoga is a popular kind of fitness all around the world. Proper posture is the cornerstone of yoga. Despite all of yoga's health benefits, bad posture can cause problems including sprains and soreness in the muscles. Over the past few years, working remotely has drawn more interest from people than working in person. Our approach is beneficial to people who are used to living online and struggle to find the time to visit yoga facilities. An increasing number of people are interested in automating the analysis of yoga postures in order to improve practitioners' performance and knowledge of the poses as a result of technological integration. Our technique is used by the suggested system to accurately extract features and estimate pose from input photos or video frames. The model classifies the yoga positions using our system's web cameras, and the image is used as input. The asana is improved by enhancing the input derived from the yoga postures through the application of multiple deep learning models.

Keywords: Deep Learning, Media Pipe, Classification, Posture Detection.

I. INTRODUCTION

Human activity recognition using computer vision has garnered significant interest from researchers due to its wide-ranging applications across domains such as robotics, human-computer interaction, gaming, video surveillance, biometric verification, sports monitoring, and health tracking[2]. Despite extensive exploration of recognition systems, detecting yoga postures remains relatively new and under-researched. The problem with yoga, however, is that, similar to other exercises, it is extremely important to practice it correctly as any incorrect posture can be useless and sometimes inconvenient during a yoga meeting. This highlights the necessity of having a teacher oversee the gathering and correct people's positions. Given that not every student approaches or provides assets to a teacher, an application based on machine reasoning might be used to identify yoga poses and provide tailored feedback to help students improve their body language.

Pose detection, a crucial aspect, employs deep learning models to analyze and interpret human body postures and movements. It encompasses key point identification, pose estimation, and tracking. Human pose estimation, a pivotal task in computer vision, involves identifying body joints or key points, aiding in understanding body language for activity recognition. This technique, available in 2D and 3D forms, has profound implications for real-time applications, particularly in robotics, autonomous driving, augmented reality, and motion tracking.

Pose grading, complementing pose detection, evaluates individual poses against target or expert poses. Misaligned or incorrect yoga poses can lead to various health issues, from acute pains to chronic conditions, making safe and accurate yoga practice essential[3,4,5]. An AI model capable of real-time yoga pose estimation and grading could mitigate the risks associated with incorrect poses, benefiting practitioners by offering corrective feedback based on expert poses' similarity.

A self-trained model that provides real-time feedback on pose accuracy, aligning it with expert demonstrations, could significantly mitigate the risks associated with incorrect poses and benefit practitioners by reducing the likelihood of injuries such as fractures, sprains, and nerve damage.

The rising number of individuals practicing yoga underscores the importance of ensuring safe and accurate execution. By leveraging computer vision techniques for real-time pose analysis and grading, practitioners can

e-ISSN: 2582-5208
International Research Journal of Modernization in Engineering Technology and Science
(Peer-Reviewed, Open Access, Fully Refereed International Journal)
Volume:06/Issue:03/March-2024 Impact Factor- 7.868 www.irjmets.com

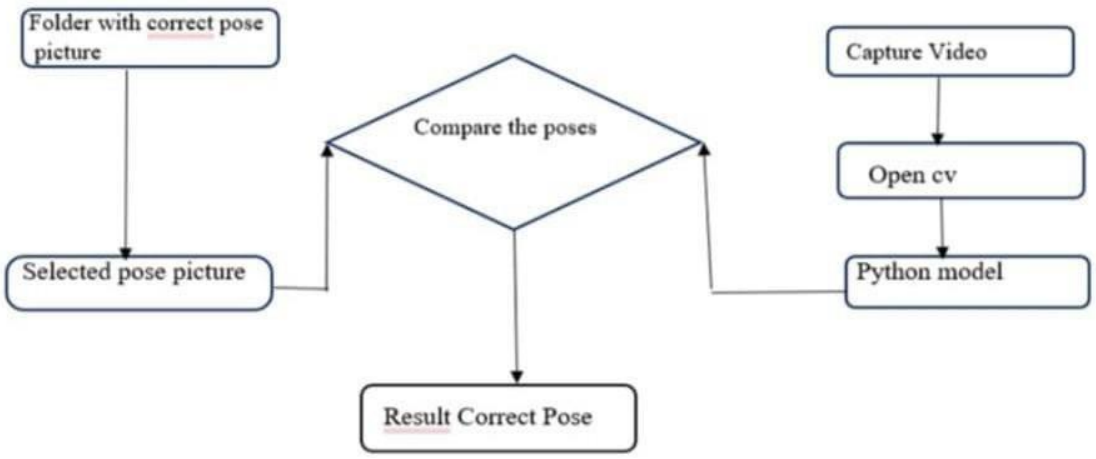
enhance their yoga experience and minimize the risk of injuries associated with incorrect poses.

This paper presents a system that uses many deep learning approaches to recognize and correct yoga positions in real time[6]. The idea is to provide in the user's home the same environment as a yoga class, where the system recognizes and corrects yoga positions automatically. It can be difficult to find an inexpensive yoga session that works with a user's schedule. By providing on-demand yoga courses with real-time feedback on the yoga program, this article aims to close this gap.

The core concept revolves around developing a deep learning model capable of accurately categorizing a user's yoga pose. This is achieved by training the model on a dataset of yoga videos, enabling it to recognize and provide corrective feedback on various poses[7,8]. Through this approach, users can experience the benefits of a guided yoga practice from the comfort of their own home, with the added advantage of receiving instant feedback to refine their technique.

II. METHODOLOGY

Our methodology expects to naturally perceive the client's Yoga asanas from ongoing and recorded recordings. a yoga identification system that uses a regular RGB camera. The data-set, which is widely accessible, was collected from 15 individuals (10 men and 5 women) utilizing HD 1080p Logitech cameras. Open Pose is used to capture the stoner on camera and provide commentary on significant elements. Since handcrafted features are not necessary for end-to-end deep literacy-grounded frames, additional asanas may be added by simply training the model with new data once more. The technique can be decayed into four fundamental advances[9]. To begin with, information assortment is performed which can either be an ongoing cycle running in corresponding with identification or can be recently recorded recordings. Second, OpenPose is utilized to distinguish the joint areas utilizing Part Confidence Maps[10] and Part Affinity Fields followed by bipartite coordinating and parsing. The essential points that have been found are fed into our model, where CNN looks for examples and LSTM analyzes how they change over time. Finally, the model and preparation plan for the surveying technique and framewise expectation on 45 edges (1.5 s) of yield are analyzed.10.1 Inversion of Pose [11]. This is the first step of our pipeline and the OpenPose library is used for it. On account of recorded recordings, this progression happens offline, though for continuous forecasts, it happens web based utilizing contribution from the camera to flexibly keypoints to the proposed model. OpenPose is an opensource library for multi-individual keypoint location, which recognizes the human body, hand, and facial keypoints together [12]. The places of 18 keypoints followed by the OpenPose, For example ears, eyes, nose, neck, shoulders, hips, knees, lower legs, elbows, and wrists. The posture extraction was performed at the default goal of OpenPose network for ideal execution.



The angle and separation between the joints are the primary factors in this paradigm that determine how the asana is performed. A NumPy function called arc tan will be used to determine the angles between the milestones of various body corridors and joints using Media pipe[13]. According to the angles, it'll estimate the posture and also recoup the instructions of that asana and it'll read out the instructions. An overall architecture of the given system is shown in illustration below.

Visuals can be obtained through various means, such as utilizing the camera module found on smartphones, which is now widely available, or using a webcam. This method is practical for capturing images since almost

e-ISSN: 2582-5208
International Research Journal of Modernization in Engineering Technology and Science
(Peer-Reviewed, Open Access, Fully Refereed International Journal)
Volume:06/Issue:03/March-2024 Impact Factor- 7.868 www.irjmets.com

everyone possesses one of these types of input devices[14]. The camera serves as the input component for the system, whether it's a webcam, a mobile camera, or a divided camera module.

This system's primary function is to handle visual data from a camera using Convolutional Neural Networks. CNNs are a subset of deep learning algorithms created especially for the analysis of visual data, such pictures or movies. CNNs are very useful for applications like image recognition and classification because they can automatically learn hierarchical features from raw pixel data through layers of convolutional filters and pooling operations[15]. The CNN model is trained on massive datasets using pictures of people in a variety of stances in order to identify poses. The model can recognize and categorize positions effectively even in difficult circumstances with varying body orientations and ambient conditions because to its intensive training, which teaches it intricate patterns and relationships between various body components.

Furthermore, the integration of MediaPipe, an open-source framework for real-time perception tasks, enhances the system's capability for pose estimation. MediaPipe provides pre-trained models and tools specifically designed for tasks like human pose detection and tracking. By leveraging MediaPipe's algorithms and capabilities, the system can efficiently extract pose information from the camera feed in real-time. This includes identifying crucial points and places on the human body, such as joints and key anatomical landmarks[16]. These points serve as essential cues for understanding body gestures and movements, enabling applications ranging from fitness tracking and augmented reality to human-computer interaction[17]. Overall, the combination of CNN-based pose identification and MediaPipe's real-time processing capabilities empowers the system to accurately analyze human poses and gestures across a wide range of scenarios, opening up opportunities for various practical applications in fields like healthcare, entertainment, and sports analysis.

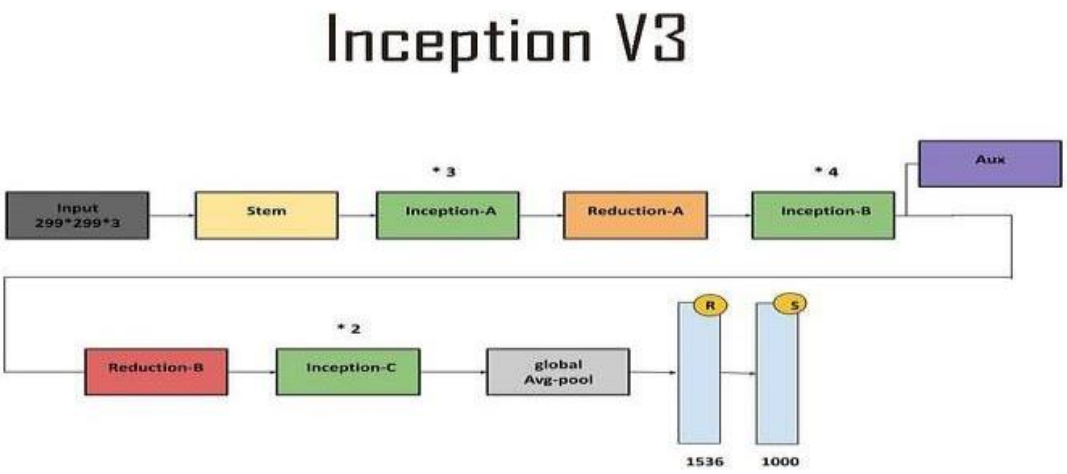
The camera receives images and feeds all the data to the model. Once the visual input is received, we trained a model using inception v3. This system is capable of recognizing a variety of poses accurately[18]. In conclusion, we maximize the utilization of datasets to their fullest extent possible. Media Pipe is utilized for pose identification, taking in the stoner's input data. Using this data, we generate an accurate representation of the stoner's posture. Key points and locations on the human body are identified through milestones in gesture recognition.

III. ALGORITHM

A. Inception v3:

As a member of the Inception family of models, Google created the deep convolutional neural network (CNN) architecture known as Inception-v3. It is renowned for its effectiveness and high accuracy and is intended for use in object identification and image classification jobs[19]. Building on earlier Inception architectures, Inception-v3 introduces a number of enhancements to boost performance and lower computational complexity.

It represents a significant advancement in deep learning for image classification tasks, offering a balance between accuracy, efficiency, and computational cost[20]. It has been widely adopted in both research and industry for various computer vision applications, including image classification, object detection, and image recognition



e-ISSN: 2582-5208
International Research Journal of Modernization in Engineering Technology and Science
(Peer-Reviewed, Open Access, Fully Refereed International Journal)
Volume:06/Issue:03/March-2024 Impact Factor- 7.868 www.irjmets.com

Incorporating Inception-v3 into yoga pose detection through transfer learning offers a powerful approach to leveraging its capabilities originally designed for image classification tasks.

While leveraging Inception-v3 as a foundational model for yoga pose detection is promising, additional refinement steps such as data augmentation, regularization techniques[21], and post-processing algorithms may be warranted to optimize performance further. Additionally, optimizing the model for real-time inference in deployment scenarios may necessitate additional considerations for efficiency and speed.

B. Opencv:

A software library for computer vision and machine learning, OpenCV is an acronym for Open Source Computer Vision Library. Originally developed by Intel in 1999, OpenCV has since become one of the most widely used libraries for computer vision tasks due to its robustness, efficiency, and extensive functionality One of the key strengths of OpenCV is its extensive collection of pre-trained models and algorithms, which allows developers to quickly build applications without needing to develop complex algorithms from scratch[22]. Additionally, OpenCV provides a modular and flexible architecture, making it easy to integrate with other libraries and frameworks. OpenCV can serve as a powerful tool for yoga pose detection, leveraging its robust computer vision capabilities and image processing functions. Although OpenCV doesn't come with pre-trained models specifically designed for yoga pose detection, it provides a versatile framework for building customized pose detection systems tailored to specific needs.

Here's a broad approach to implementing yoga pose detection using OpenCV:

a. **Data Collection:** Start by assembling a dataset comprising yoga images or videos showcasing a diverse range of poses. Ensure the dataset encompasses various angles, lighting conditions, and individuals to capture the nuances of different poses accurately.

b. **Preprocessing:** Enhance the quality of the images or frames through preprocessing techniques like resizing, grayscale conversion, noise reduction, and contrast adjustment. These steps help improve the accuracy of subsequent pose detection algorithms.

c. **Pose Estimation:** Develop a pose estimation algorithm to detect key points or joints in the body. OpenCV offers various algorithms and techniques for keypoint detection, such as:

- **Histogram of Oriented Gradients (HOG):** Compute HOG descriptors to identify human figures in images.
- **Haar Cascades:** Utilize pre-trained Haar cascades to detect specific body parts like the face, hands, or legs.
- **Deep Learning Models:** Employ deep learning-based pose estimation models like OpenPose, which leverages Convolutional Neural Networks (CNNs) to detect keypoints accurately.

d. **Pose Classification:** Once key points are detected, classify the yoga pose based on the spatial arrangement of these keypoints. This step may involve techniques such as template matching, feature extraction, or machine learning classifiers to determine the pose accurately.

e. **Feedback and Correction:** Provide real-time feedback to users on their detected yoga poses. This feedback can take the form of visual cues, such as highlighting incorrect postures or providing audio instructions for corrections, enabling users to adjust their poses accordingly.

f. **Integration with GUI:** Develop a user-friendly graphical interface using OpenCV or other libraries like Tkinter (for Python) to facilitate interaction with the pose detection system. This GUI can display real-time feedback and allow users to navigate through different poses seamlessly.

g. **Testing and Evolution:** Thoroughly test the pose detection system using diverse datasets and evaluate its performance in terms of accuracy, speed, and robustness. Fine-tune the system as needed to address any shortcomings and ensure optimal performance.

C. Mediapipe:

MediaPipe is an open-source framework developed by Google that offers a comprehensive set of tools and algorithms for building machine learning-based applications in various domains, with a primary focus on computer vision and media processing tasks. Launched in 2019, MediaPipe simplifies the development of real-time perception and understanding systems by providing a flexible and scalable platform[23].

At its core, MediaPipe enables developers to create pipelines for processing and analyzing media inputs such as images, videos, and audio streams. These pipelines consist of modular components called "graphs," each

e-ISSN: 2582-5208
International Research Journal of Modernization in Engineering Technology and Science
(Peer-Reviewed, Open Access, Fully Refereed International Journal)
Volume:06/Issue:03/March-2024 Impact Factor- 7.868 www.irjmets.com

performing specific tasks such as object detection, pose estimation, hand tracking, facial recognition, and more. Developers can easily assemble these components to construct complex multimedia processing pipelines tailored to their application requirements.

One of the key features of MediaPipe is its support for hardware acceleration, enabling efficient execution of computationally intensive tasks on various platforms, including CPUs, GPUs, and specialized hardware like Google's Edge TPU[24]. This ensures optimal performance and scalability across a wide range of devices, from mobile phones and laptops to edge devices and IoT (Internet of Things) devices.

MediaPipe presents a robust solution for yoga pose detection, capitalizing on its real-time perception abilities and comprehension of human body movements. Below outlines how MediaPipe can be applied to yoga pose detection:

1. **Pose Detection Module:** MediaPipe offers a pre-configured pose detection module capable of accurately gauging the keypoints and skeletal structure of an individual's body in real-time from video inputs. This module relies on deep learning techniques to identify and track key body joints like the shoulders, elbows, wrists, hips, knees, and ankles.
2. **Customization:** Although MediaPipe provides a pre-trained pose detection model, developers retain the freedom to fine-tune or tailor the model to meet specific criteria. This involves refining the model through training on a dataset of yoga poses, enhancing its precision and adaptability for recognizing yoga-specific poses.
3. **Integration with Yoga Pose Dataset:** Yoga photos or videos showing different positions from different perspectives can be compiled by developers into a dataset. The performance of the posture identification model in identifying yoga poses is trained and assessed using this dataset.
4. **Real Time Feedback:** Once deployed, the pose detection model delivers real-time feedback to users during yoga practice sessions. Feedback mechanisms may encompass visual cues such as highlighting correct or incorrect posture alignments, textual instructions for corrections, or generating auditory feedback for guided yoga practice.
5. **Integration with Other Modules:** MediaPipe facilitates seamless integration of the pose detection module with additional modules or components to extend functionality. For instance, developers can integrate pose detection with audio processing modules to provide guided yoga instructions based on identified poses, or merge it with augmented reality (AR) components for immersive yoga experiences.
6. **Performance Optimization**MediaPipe is optimized for superior performance and efficiency, ensuring smooth and responsive real-time pose detection even on devices with limited resources like smartphones or tablets. Leveraging MediaPipe's hardware acceleration capabilities enables developers to achieve peak performance across various platforms.

Essentially, MediaPipe is a flexible way to construct advanced pose detection systems for yoga, allowing programmers to design engaging and dynamic experiences that improve users' yoga practice trips.

IV. CONCLUSION

Our system is a culmination of meticulous design, incorporating a multifaceted approach to enhancing yoga practice. At its core, the system comprises a sophisticated channel for disguise identification, allowing for the precise recognition of various yoga poses. Coupled with this is the capability for precise point localization on the human body, enabling detailed analysis of posture and alignment during yoga sessions[25]. Through these components, our technique aims not only to assist individuals in practicing yoga correctly on their own but also to mitigate potential ailments that may arise from improper poses, promoting overall well-being and safety.

Central to our approach is the integration of an error identification method, which serves as a guiding light for practitioners seeking to refine their yoga techniques. By pinpointing areas of misalignment or improper posture, the system empowers individuals to make informed adjustments, reducing the risk of injuries and discomfort. Leveraging deep literacy techniques, our system provides personalized feedback tailored to the practitioner's unique needs, fostering continuous improvement in yoga practice. Moreover, the system's design is optimized for evaluating yoga poses from a frontal perspective, ensuring comprehensive assessment and feedback. Seamlessly integrated into a user-friendly dashboard, our model caters to the needs of practitioners,

offering intuitive navigation and valuable insights to support their journey towards mastery in yoga.

V. REFERENCES

[1] Kumar, Deepak & Sinha, Anurag. (2020). Yoga Pose Detection and Classification Using Deep Learning. International Journal of Scientific Research in Computer Science Engineering and Information Technology.10.32628/CSEIT206623.

[2] Gong, W.; Zhang, X.; Gonzàlez, J.; Sobral, A.; Bouwmans, T.; Tu, C.; Zahzah, E.H. Human pose estimation from monocular images: A comprehensive survey. Sensors (2016), 16, 1966.

[3] Satyam Goyal And Animesh Jain. Yoga pose perfection using Deep learning: An Algorithm to Estimate the Error in yogic Poses. American High school, Fremant, CA, USA(2021).

[4] A. Agarwal and B. Triggs, "3D human pose from silhouettes by relevance vector regression", Intl Conf. on Computer Vision & Pattern Recogn.pp.882–888, 2004.

[5] M. Li, Z. Zhou, J. Li and X. Liu, "Bottom-up pose estimation of multiple person with bounding box constraint", 24th Intl. Conf. Pattern Recogn.,(2018).

[6] S. Kreiss, L. Bertoni, and A. Alahi, "PifPaf: composite fields for human pose estimation", IEEE Conf. Computer Vision and Pattern Recogn, (2019).

[7] Kumarapu, L.; Mukherjee, P. Animepose: Multi-person 3d pose estimation and animation. Pattern Recognit. Lett. (2021), 147, 16–24.

[8] Pardos, A.; Menychtas, A.; Maglogiannis, I. On unifying deep learning and edge computing for human motion analysis in exergames development. Neural Comput. Appl. (2022), 34, 951–967.

[9] Gamra, M.B.; Akhloufi, M.A. A review of deep learning techniques for 2D and 3D human pose estimation. Image Vis. Comput. (2021), 114, 104282.

[10] Bin, Y.; Chen, Z.M.; Wei, X.S.; Chen, X.; Gao, C.; Sang, N. Structure-aware human pose estimation with graph convolutional networks. Pattern Recognit. (2020), 106, 107410.

[11] Alsubait, T.; Sindi, T.; Alhakami, H. Classification of the Human Protein Atlas Single Cell Using Deep Learning. Appl. Sci. (2022), 12, 11587.

[12] Dubey, S.; Dixit, M. A comprehensive survey on human pose estimation approaches. Multimed. Syst. (2023), 29, 167–195.

[13] Dang, Q.; Yin, J.; Wang, B.; Zheng, W. Deep learning based 2D human pose estimation: A survey. Tsinghua Sci. Technol. (2019), 24,663–676.

[14] Song, L.; Yu, G.; Yuan, J.; Liu, Z. Human pose estimation and its application to action recognition: A survey. J. Vis. Commun. Image Represent. (2021), 76, 103055.

[15] Chen, Y.; Tian, Y.; He, M. Monocular human pose estimation: A survey of deep learning-based methods. Comput. Vis. Image Underst.(2020), 192, 102897.

[16] Zheng, C.; Wu, W.; Chen, C.; Yang, T.; Zhu, S.; Shen, J.; Kehtarnavaz, N.; Shah, M. Deep Learning- Based Human Pose Estimation: A Survey. J. ACM (2023), 37, 35.

[17] Badiola-Bengoa, A.; Mendez-Zorrilla, A. A Systematic Review of the Application of Camera-Based Human Pose Estimation in the Field of Sport and Physical Exercise. Sensors (2021), 21, 5996.

[18] Desai, M.M.; Mewada, H.K. Review on Human Pose Estimation and Human Body Joints Localization. Int. J. Comput. Digit. Syst. (2021), 10, 883–898.

[19] Luvizon, D.C.; Tabia, H.; Picard, D. Human pose regression by combining indirect part detection and contextual information. Comput. Graph. (2019), 85, 15–22.

[20] Jiao, Y.; Chen, H.; Feng, R.; Chen, H.; Wu, S.; Yin, Y.; Liu, Z. GLPose: Global-Local Representation Learning for Human Pose Estimation. ACM Trans. Multimed. Comput. Commun. Appl. (2022), 18, 1–16.

[21] Chen, H.; Feng, R.; Wu, S.; Xu, H.; Zhou, F.; Liu, Z. 2D Human pose estimation: A survey. Multimed. Syst. (2022), 29, 3115–3138.

[22] Xu, D.; Zhang, R.; Guo, L.; Feng, C.; Gao, S. LDNet: Lightweight dynamic convolution network for human pose estimation. Adv. Eng. Inform. (2022), 54, 101785.

[23] Du, C.; Yan, Z.; Yu, H.; Yu, L.; Xiong, Z. Hierarchical Associative Encoding and Decoding for Bottom- Up

Human Pose Estimation.IEEE Trans. Circuits Syst. Video Technol. (2022), 33, 1762–1775.

[24] Zhao, Y.; Luo, Z.; Quan, C.; Liu, D.; Wang, G. Cluster-wise learning network for multi-person pose estimation. Pattern Recognit. (2020), 98, 107074.

[25] Nasr, M.; Ayman, H.; Ebrahim, N.; Osama, R.; Mosaad, N.; Mounir, A. Realtime multi-person 2D pose estimation. Int. J. Adv. Netw. Appl. (2020), 11, 4501–4508.